Ex:No: 1E Date: 28.01.26	Monthly Sales Analysis Using Vectors

Problem Description:

A retail store records its monthly sales figures in a vector. Write an R program to analyze the sales data by calculating the total sales, average monthly sales, highest sales value, lowest sales value, and identifying months with sales above the average.

Objective:

To apply vector operations in R for storing and analyzing numerical business data such as monthly sales figures.

Sample Input	Expected Output
Enter number of months: 5 Enter sales for month 1 : 20000 Enter sales for month 2 : 22000 Enter sales for month 3 : 19000 Enter sales for month 4 : 19500 Enter sales for month 5 : 21000	Sales Vector: 20000 22000 19000 19500 21000 Total Sales: 101500 Average Sales: 20300 Highest Sales: 22000 Lowest Sales: 19000 Above Average Sales: 22000 21000

Program:

```
# Monthly Sales Analysis using Vectors
# User input n and sales
n <- as.integer(readline("Enter number of months: "))
sales <- c()
for(i in 1:n) {
    value <- as.numeric(readline(prompt = paste("Enter
                                 sales for month", i, ": ")))
    sales <- c(sales, value)
}

# Display vector

cat("\n Sales vector : ", sales, "\n")
```

SAMPLE INPUT:

Enter number of months: 4

Enter sales for month 1: 15000

Enter sales for month 2: 18000

Enter sales for month 3: 17000

Enter sales for month 4: 20000

EXPECTED OUTPUT:

Sales Vector: 15000 18000 17000 20000

Total Sales: 70000

Average Sales: 17500

Highest Sales: 20000

Lowest Sales: 15000

Above Average Sales: 18000 20000

```
for (i in 1:n){
    cat("Enter sales for month", i, ":")
    sales[i]<-as.numeric
        (readline())
```

Calculations

```
total_sales <- sum(sales)
average_sales <- mean(sales)
highest_sales <- max(sales)
lowest_sales <- min(sales)

cat("Total sales: ", total_sales, "\n")
cat("Average sales: ", average_sales, "\n")
cat("Highest sales: ", highest_sales, "\n")
cat("Lowest sales: ", lowest_sales, "\n")
```

Above average sales

```
above_avg <- sales[sales > average_sales]
cat("Above Average Sales: ", above_avg, "\n")
```

Result:

Thus, the program analyzes monthly sales data using vector operations in R. It demonstrates efficient aggregation and comparison of business data using vectors.